Submission PY1-MID-SUB02

Question PY1-MID-Q01
work / setup:
1. range(4) gives 0 through 3
2. keep even i: 0, 2
crossed-out arithmetic: [smudged]
FINAL (maybe): [0, 4] plus 1

Question PY1-MID-Q02
work / setup:
1. counts = Counter(xs)
2. for x in xs: if counts[x] == 1: return x
crossed-out arithmetic: [smudged]
FINAL (maybe): scan counts, then scan in order; return first count==1 else None; O(n) time, O(n) space plus 1

Question PY1-MID-Q03
work / setup:
1. A single pass fixes only adjacent inversions encountered once.
crossed-out arithmetic: [smudged]
FINAL (maybe): repeat passes or track the shrinking unsorted suffix; one pass alone is not a full sort plus 1

Question PY1-MID-Q04
work / setup:
1. linear scan: up to 64 comparisons, $\Theta(n)$
crossed-out arithmetic: [smudged]
FINAL (maybe): linear $\Theta(n)$, up to 64; binary $\Theta(\log n)$, about 7 plus 1

Question PY1-MID-Q05
work / setup:
1. Removing shifts later elements left while the iterator advances.
crossed-out arithmetic: [smudged]
FINAL (maybe): mutation shifts elements and skips checks; use a filtered copy or iterate over a copy plus 1

Question PY1-MID-Q06
work / setup:
1. range(4) gives 0 through 3
2. keep even i: 0, 2
crossed-out arithmetic: [smudged]
FINAL (maybe): [0, 4] plus 1

Question PY1-MID-Q07
work / setup:
1. counts = Counter(xs)

2. for x in xs: if counts[x] == 1: return x
crossed-out arithmetic: [smudged]
FINAL (maybe): scan counts, then scan in order; return first count==1 else None; O(n) time, O(n) space plus 1

Question PY1-MID-Q08
work / setup:
1. A single pass fixes only adjacent inversions encountered once.
crossed-out arithmetic: [smudged]
FINAL (maybe): repeat passes or track the shrinking unsorted suffix; one pass alone is not a full sort plus 1